AWS S3 (Simple Storage Service) - SAA-C03 Study Notes

1. S3 Overview

- **Type:** Object Storage (not Block or File storage).
- **Structure:**
  - **Buckets:** Root level containers. Must have a **globally unique name** (across all AWS regions/accounts).
  - **Objects:** Files stored in buckets.
  - **Keys:** The full path to the object (e.g., `images/logo.png`).
- **Size:** Unlimited total storage. Single object size from 0 bytes to **5 TB**.
- **Durability:** 99.999999999% (11 9s) for Standard classes.
- **Consistency Model:** Strong consistency for all GET, PUT, and LIST operations (read-after-write consistency).

2. Security: IAM vs. Bucket Policies

S3 security is a combination of User-Based policies and Resource-Based policies.

IAM Policies (User-Based)

- Attached to IAM Users, Groups, or Roles.
- Defines what an IAM principal can do (e.g., "User A can read from Bucket X").
- Best for internal users within your AWS account.

S3 Bucket Policies (Resource-Based)

- Attached directly to the S3 Bucket.
- Defines *who* can access *this specific bucket*.
- JSON format. Key elements: `Principal`, `Effect`, `Action`, `Resource`.
- **Use Cases:**
  - Granting public access to a bucket.
  - Granting cross-account access.
  - Enforcing upload of encrypted objects only.

Evaluation Logic

- **Union of Policies:** Access is the union of IAM permissions and Bucket Policy permissions.
- **Explicit DENY:** Always overrides an ALLOW.
- **Implicit DENY:** If not explicitly allowed, it is denied.

Block Public Access

- An account-level or bucket-level setting that overrides any bucket policy to prevent accidental public exposure.
- Should be enabled unless you specifically intend to host a public static website.

3. S3 Static Website Hosting

- S3 can host static websites (HTML, CSS, JS, Images). It cannot run server-side code (PHP, Python, etc.).
- **URL Format:** `http://<bucket-name>.s3-website-<region>.amazonaws.com`
- **Requirements:**
  1. Enable "Static website hosting" in properties.
  2. Set an **Index Document** (e.g., `index.html`) and **Error Document** (e.g., `error.html`).
  3. Disable "Block Public Access".
  4. Add a **Bucket Policy** to allow `s3:GetObject` for `Principal: "*"`.
- **Exam Tip:** S3 Website endpoints do **not** support HTTPS. To get HTTPS, you must use CloudFront in front of S3.

4. S3 Versioning

- Enabled at the bucket level. Once enabled, it can only be suspended, not disabled.
- **Function:** Keeps multiple variants of an object in the same bucket.
- **Benefits:**
  - Protects against unintended overwrites.
  - Protects against accidental deletions (a "Delete Marker" is added instead of removing the file).
  - Easy rollback to previous versions.
- **MFA Delete:** Forces Multi-Factor Authentication to change the versioning state or permanently delete an object version. (Must be enabled via CLI, not Console).

5. S3 Replication

To replicate data, **Versioning must be enabled** on both the Source and Destination buckets.

- **CRR (Cross-Region Replication):**
  - Replicate to a different AWS Region.
  - Use cases: Compliance, lower latency for global users, disaster recovery.
- **SRR (Same-Region Replication):**
  - Replicate to the same AWS Region.
  - Use cases: Log aggregation, replication between production and test environments.

- **Key Notes:**
  - Only *new* objects are replicated after enabling. (Existing objects need "S3 Batch Replication").
  - Delete Markers are replicated? Configurable.
  - Permanent deletions? NOT replicated (security feature).
  - Requires an IAM Role for S3 to perform the replication.
  - **No Chaining:** If Bucket A replicates to Bucket B, and Bucket B replicates to Bucket C, objects uploaded to Bucket A are **not** replicated to Bucket C.

6. S3 Storage Classes

S3 Standard – General Purpose

- **Availability:** 99.99%
- Used for frequently accessed data.
- Low latency and high throughput.
- Sustain 2 concurrent facility failures.
- **Use Cases:** Big Data analytics, mobile & gaming applications, content distribution.

S3 Storage Classes – Infrequent Access

For data that is less frequently accessed, but requires rapid access when needed. Lower cost than S3 Standard.

- **Amazon S3 Standard-Infrequent Access (S3 Standard-IA)**
  - **Availability:** 99.9%
  - **Use cases:** Disaster Recovery, backups.
- **Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)**
  - **Durability:** High durability (99.999999999%) in a single AZ; **data lost when AZ is destroyed**
  - **Availability:** 99.5%
  - **Use Cases:** Storing secondary backup copies of on-premises data, or data you can recreate.

Amazon S3 Glacier Storage Classes

Low-cost object storage meant for archiving / backup. **Pricing:** Price for storage + object retrieval cost.

- **Amazon S3 Glacier Instant Retrieval**
  - Millisecond retrieval, great for data accessed once a quarter.
  - Minimum storage duration of 90 days.
- **Amazon S3 Glacier Flexible Retrieval** (formerly Amazon S3 Glacier)

    - **Retrieval options:** Expedited (1 to 5 minutes), Standard (3 to 5 hours), Bulk (5 to 12 hours) – free.
    - Minimum storage duration of 90 days.
- **Amazon S3 Glacier Deep Archive** – For long term storage
    - **Retrieval options:** Standard (12 hours), Bulk (48 hours).
    - Minimum storage duration of 180 days.

S3 Intelligent-Tiering

- Small monthly monitoring and auto-tiering fee.
- Moves objects automatically between Access Tiers based on usage.
- **There are no retrieval charges in S3 Intelligent-Tiering.**
- **Tiers:**
    - **Frequent Access tier (automatic):** Default tier.
    - **Infrequent Access tier (automatic):** Objects not accessed for 30 days.
    - **Archive Instant Access tier (automatic):** Objects not accessed for 90 days.
    - **Archive Access tier (optional):** Configurable from 90 days to 700+ days.
    - **Deep Archive Access tier (optional):** Configurable from 180 days to 700+ days.

7. S3 Express One Zone

- **Newest Storage Class:** High-performance, single-availability zone.
- **Structure:** Uses **Directory Buckets** (different from standard general-purpose buckets).
- **Performance:** Delivers single-digit millisecond latency.
- **Cost:** 50% lower request costs compared to S3 Standard.
- **Use Case:** frequently accessed data, AI/ML training, financial modeling, media processing.
- **Durability:** Data is stored in a single AZ; if that AZ is destroyed, data is lost (unlike Regional classes).